LOCMARCA 2026 · COURSE 1

Observation products

In situ vs satellite, and what a gridded dataset really tells you

UNA, Heredia, Costa Rica · Tuesday 17 November 2026

GOALS

What you should take away

1

Choose

Pick the right product for a question: variable, depth, period, resolution.

2

Understand

Know how the numbers were made: measured, interpolated, or averaged.

3

Question

Spot the limits of a dataset before they turn into wrong conclusions.

Not a catalogue: a way of reading **any** dataset you will meet this week.

ROADMAP

Six questions, one hour

1

In situ vs satellite

Who measures what, and how

2

Ocean, atmosphere, BGC

The variables and their products

3

2D vs 3D

The surface and the water column

4

Coverage

How often, how fine, how long: advantages and limits

5

Climatology vs interannual

The typical year and the real ones

6

Gridded data

What interpolation adds, and what it hides

PART 1 · IN SITU VS SATELLITE

Every dataset is a compromise between what is measured, where, when, and how precisely.

No product does everything. Knowing which compromise it made is most of the skill.

PART 1 · IN SITU VS SATELLITE

In situ: measured where the instrument is

Ships

CTD casts and water samples: full depth, physics and biogeochemistry, but rare and expensive

Moorings

e.g. the TAO/TRITON array in the tropical Pacific: one point, measured continuously for years

Argo floats

About 4,000 floats worldwide, a temperature and salinity profile every 10 days, 0–2,000 m

And also

Surface drifters, gliders, tide gauges, coastal and weather stations

Strengths

Direct and accurate
Below the surface, down to the bottom
Long records at fixed points
The reference used to calibrate satellites

Limits

Sparse and irregular in space
Gaps in time
Costly: few in coastal and developing regions

Satellites: the whole surface, from above

What	Sensor	The catch
Sea surface temperature	Infrared and microwave radiometers	Infrared is blocked by clouds; microwave sees through them but is coarse (tens of km)
Sea level anomaly	Altimeters	Measured along tracks only; gridded maps smooth out small eddies
Chlorophyll	Ocean colour radiometers	Clouds; sees only the first metres of the water
Surface wind	Scatterometers	Less reliable close to the coast
Sea surface salinity	SMOS, SMAP	Coarse (≈ 40–50 km) and noisy near land

Global, repeated, consistent — but **only the surface**, measured **indirectly**, and always calibrated against in situ data.

Side by side

	In situ	Satellite
Depth	From the surface to the bottom	Surface only (skin to a few metres)
Coverage	Sparse points, irregular	Global, every 1 to 10 days
Accuracy	High: a direct measurement	Indirect: needs an algorithm and calibration
Gaps	Wherever nobody went	Clouds, coasts, between tracks
Records	Ships since the late 1800s; Argo since 2000	SST from 1981, altimetry 1993, ocean colour 1997

PART 1 · IN SITU VS SATELLITE

Neither is enough alone: we combine them

In situ

Accurate, deep, but sparse

Satellite

Dense, global, but surface only

→

Combine

Interpolation (objective analysis) or data assimilation in a model

→

Gridded products

Gap-free maps (L4), climatologies, reanalyses

This week: **OSTIA** SST (L4) and **WOA** climatology (TP1b) · **ARMOR3D** 3D observations, **GLORYS12** ocean reanalysis and **Wind L4** (TP2a, TP2b) · **GlobColour** chlorophyll and the **PISCES** model (TP2c) · **CROCO** regional model (TP3)

PART 2 · OCEAN, ATMOSPHERE, BGC

Three domains, three sets of variables

Domain	Key variables	Observed by	Used this week
Ocean physics	Temperature, salinity, sea level, currents	Argo, CTD, moorings, altimetry, SST radiometers	OSTIA SST, WOA, ARMOR3D, GLORYS12 reanalysis
Atmosphere	Wind, air temperature, pressure, rain, radiation	Weather stations, radiosondes, scatterometers	Wind L4: ERA5 corrected by scatterometers
Biogeochemistry	Chlorophyll, oxygen, nutrients, pH	Ocean colour, bottle samples, BGC-Argo floats	GlobColour chlorophyll, PISCES model (TP2c)

Biogeochemistry is the **least observed**: few in situ samples, and satellites only see surface chlorophyll.

PART 2 · OCEAN, ATMOSPHERE, BGC

One process, three domains: the Papagayo jet

ATMOSPHERE

November to March, the trade winds funnel through a gap in the mountains and blow as a strong jet over the Gulf of Papagayo.

Seen by: scatterometers, ERA5 (TP2a)

→

OCEAN

The wind mixes and pushes surface water offshore; cold water rises. SST falls to ≈ 24 °C in February 2023.

Seen by: OSTIA SST (TP1a, TP1b, TP2a)

→

BIOGEOCHEMISTRY

Nutrients reach the sunlit layer: phytoplankton blooms and surface chlorophyll rises.

Seen by: ocean colour (TP2c)

Understanding a process often needs products from **all three** domains: that is the spirit of the TP4 mini-project.

PART 3 · 2D VS 3D

2D or 3D: is there a depth axis?

2D The surface

`(time, latitude, longitude)`

Satellite SST, sea level, chlorophyll, 10 m wind

TP1b: analysed_sst(time, latitude, longitude)

3D The water column

`(time, depth, latitude, longitude)`

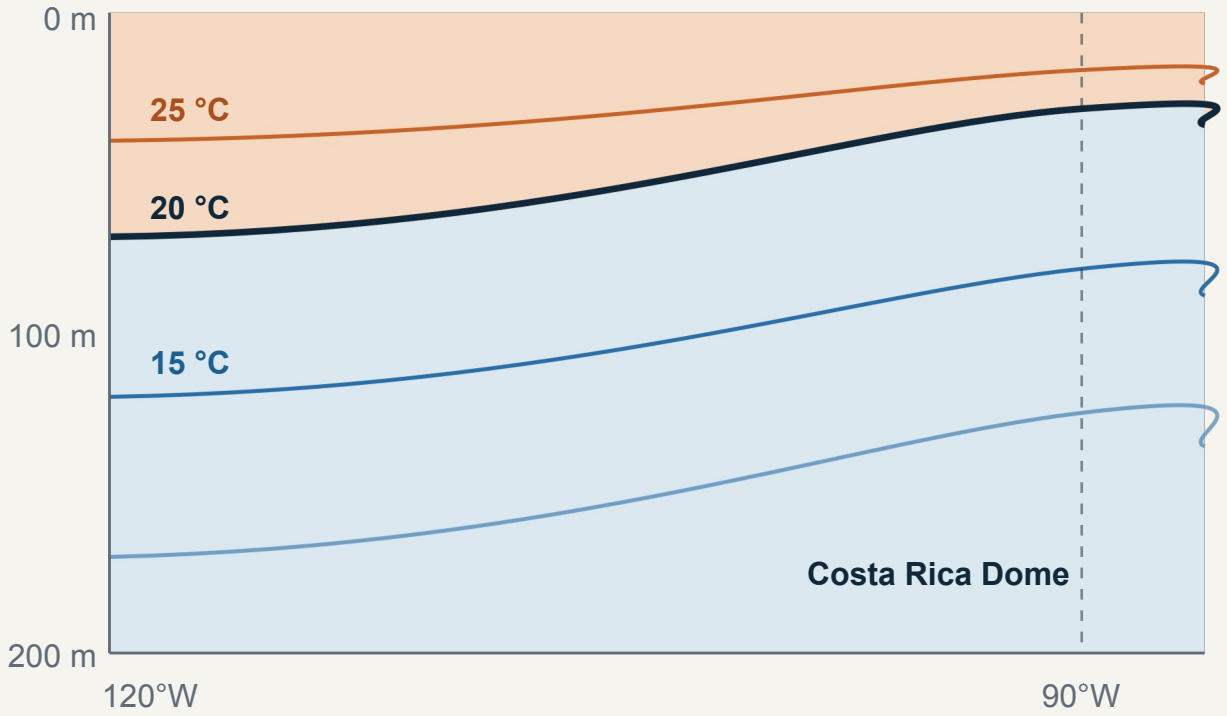
Argo and CTD profiles, WOA, ocean reanalyses, model outputs

TP1b: temperature(T, Z, Y, X)

3D is much heavier: one month of daily 3D ocean data (≈ 165 MB) outweighs a whole year of daily SST (≈ 42 MB) on the same region. **Select before you download** (TP2a).

PART 3 · 2D VS 3D

Why depth matters: the Costa Rica Dome



The thermocline rises from ≈ 70 m to ≈ 30 m near 90°W, 9°N.

A satellite only sees the **surface signature**: cooler SST, more chlorophyll.

To see the **structure** you need 3D data: Argo, CTD, WOA, a reanalysis or a model.

Schematic. You will draw the real section from WOA in TP1b.

PART 4 · COVERAGE IN SPACE AND TIME

How fine, how often, how deep, how long

Platform	Spatial sampling	Time sampling	Depth	Since
Argo floats	≈ one float per 3° × 3°	A profile every 10 days	0–2,000 m	2000
Moorings	A few fixed points	Minutes to hours	Surface to 500 m	1980s
Ship surveys	Sparse lines	Occasional campaigns	To the bottom	Late 1800s
Infrared SST	≈ 1 km	Daily, if no clouds	Surface	1981
Altimetry	Along tracks; maps ≈ 1/8°–1/4°	≈ 10 days per track	Surface	1993
Ocean colour	≈ 1–4 km	Daily, if no clouds	First metres	1997
ERA5 reanalysis	≈ 31 km	Hourly	Atmosphere	1940

Orders of magnitude to compare platforms: check each product's documentation for exact values.

PART 5 · CLIMATOLOGY VS INTERANNUAL

The typical year and the real ones

Climatology

The average of many years (often 30) for each month or season: the **typical year**. Examples: WOA (TP1b), GLORYS12 1993–2023 (TP2b).

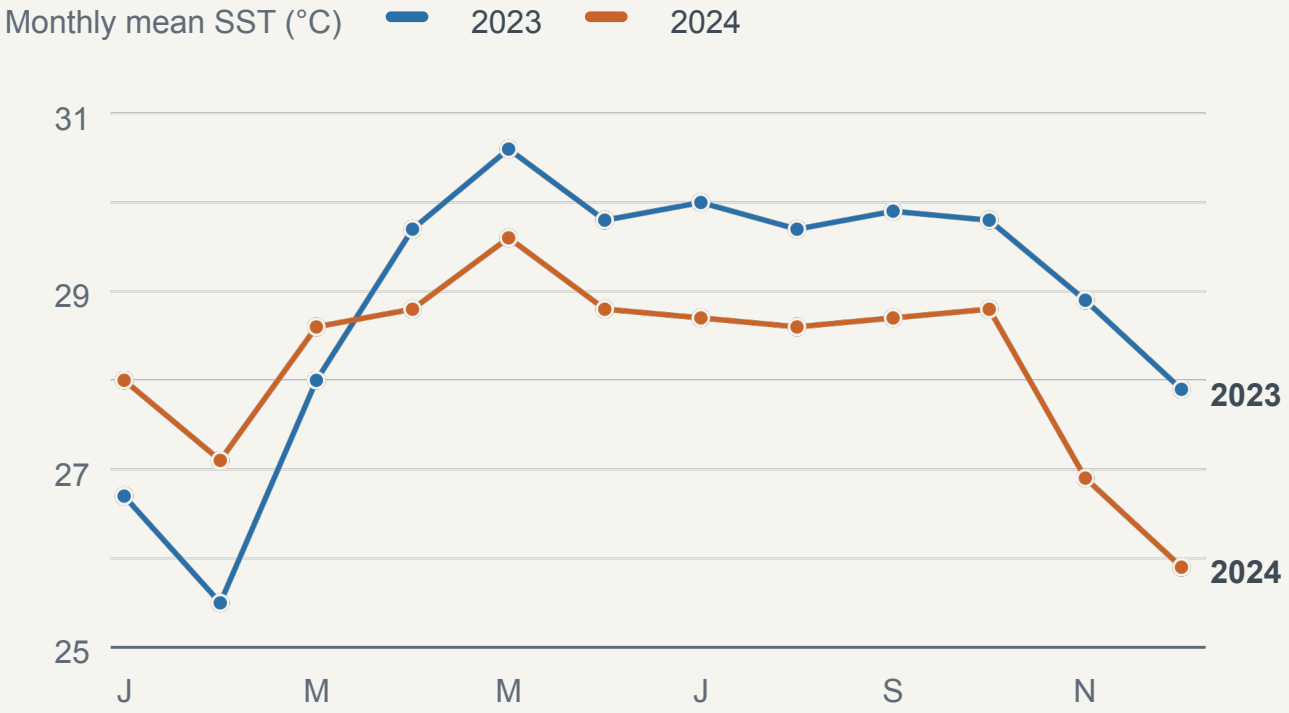
Interannual variability

How one year differs from another. In the eastern tropical Pacific the main driver is **El Niño / La Niña** (ENSO).

anomaly = observation – climatology Removes the seasonal cycle, keeps what is unusual.

PART 5 · CLIMATOLOGY VS INTERANNUAL

Same seasonal cycle, different years



OSTIA SST at 87°W, 10°N (Gulf of Papagayo)

Both years: cold Jan–Mar, warm May–Oct. That shape is the **seasonal cycle**.

Jan–Feb 2024: **+1.3 to +1.6 °C** (end of El Niño 2023–24). Nov–Dec 2024: **−2 °C**.

That offset is **interannual variability**. Two years are not a climatology: you need ~30.

PART 5 · CLIMATOLOGY VS INTERANNUAL

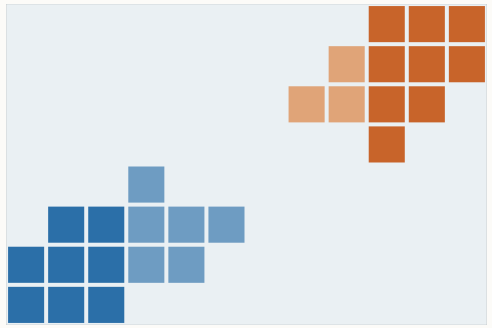
Know what kind of time axis you have

Product	Time axis	Good for
WOA 2009	4 seasons, long-term mean	The typical state; a reference for anomalies
OSTIA SST 2023	Daily, one year	Events and one specific year
GLORYS12 reanalysis	Monthly, 1993 to today	Your own climatology and anomalies (TP2b)
Climatological model run	A "typical" year, repeated	Processes, not a given date (Course 2)

⚠ **Pitfall:** comparing a model climatology with one El Niño year mixes two things, model error and interannual variability.

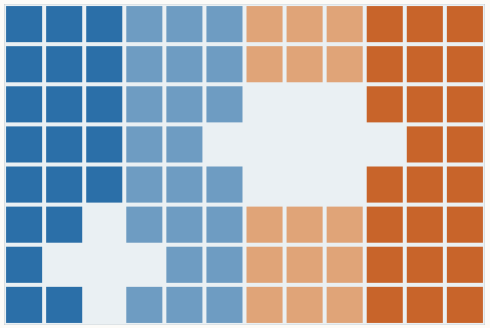
PART 6 · GRIDDED DATA

From swath to map: processing levels



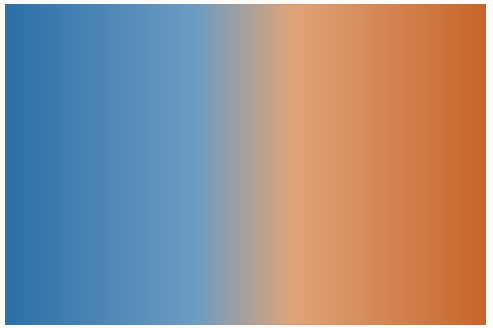
L2 · swath

Measurements along the satellite's path, at the instrument's resolution. Holes under clouds.



L3 · gridded

L2 data put on a regular grid and averaged over a day or more. Gaps remain.



L4 · gap-free

Gaps filled by interpolation, often merging several sensors. Complete and smooth. OSTIA is L4.

PART 6 · GRIDDED DATA

What interpolation gives, and what it hides

It gives you

- A value at every grid point
- Maps that are easy to plot, average and compare
- No gaps to manage in your code

It costs you

- Smoothing: small, short features fade away
- Filled values are estimates, not measurements
- Where data are scarce, the map falls back on its first guess, often a climatology
- Artefacts near coasts and islands

Look for the error or quality field shipped with the product (OSTIA: **analysis_error**) and read its documentation.

PART 6 · GRIDDED DATA

A 0.05° grid is not a 5 km resolution.

The grid spacing is the producer's choice. The **effective resolution** depends on how many observations went in: under persistent clouds, a "fine" L4 map is mostly interpolation.

TAKE-AWAY

Five questions to ask any dataset

1. **What exactly is measured?** Variable, units, depth. OSTIA gives a *foundation* SST, not the skin.
2. **Where and when are the real observations?** And where is it interpolated?
3. **What is the effective resolution?** Not just the grid spacing.
4. **A climatology or a real period?** Which years, which time step?
5. **How uncertain is it?** Error fields, quality flags, documentation.

NEXT

This week, you will use all of this

Course 0a

Anatomy of a netCDF file: where the answers to the five questions are written

TP1a, TP1b

xarray basics, then OSTIA SST and the WOA climatology on Kabre: maps, series, sections

TP2a

Download your own data from Copernicus: SST L4, 3D temperature, reanalyses

TP2b, TP2c

Climatology vs interannual variability, then biogeochemistry (chlorophyll, nitrate, oxygen)

TP3, TP4

NEMO and CROCO model outputs, then compare observations, reanalyses and models on a region of your choice

Questions?